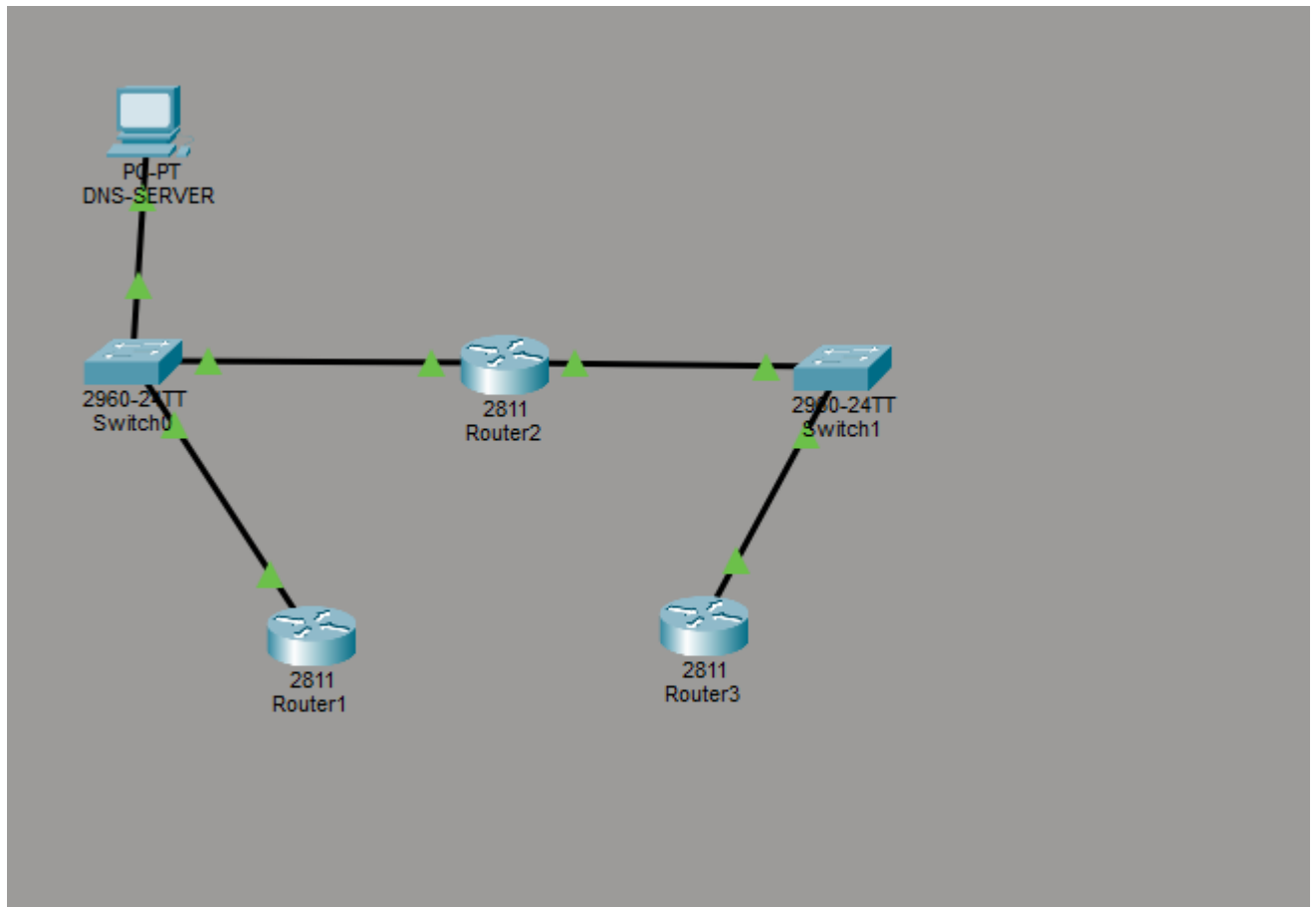


1.) Screenshot of Complete topology:



2.) Screenshot of router pinging a hostname.

```
Router3
Physical Config CLI Attributes
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 10.10.20.2 255.0.0.0
Router(config-if)#ip address 10.10.20.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
Router(config-if)#
%LINK-3-CHANGED: Interface FastEthernet0/1, changed state to administratively down
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#configure terminal
Invalid input detected at '^' marker.
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip domain-lookup
Router(config)#name-server 10.10.10.10
Invalid input detected at '^' marker.
```

Router3

Physical Config CLI Attributes

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

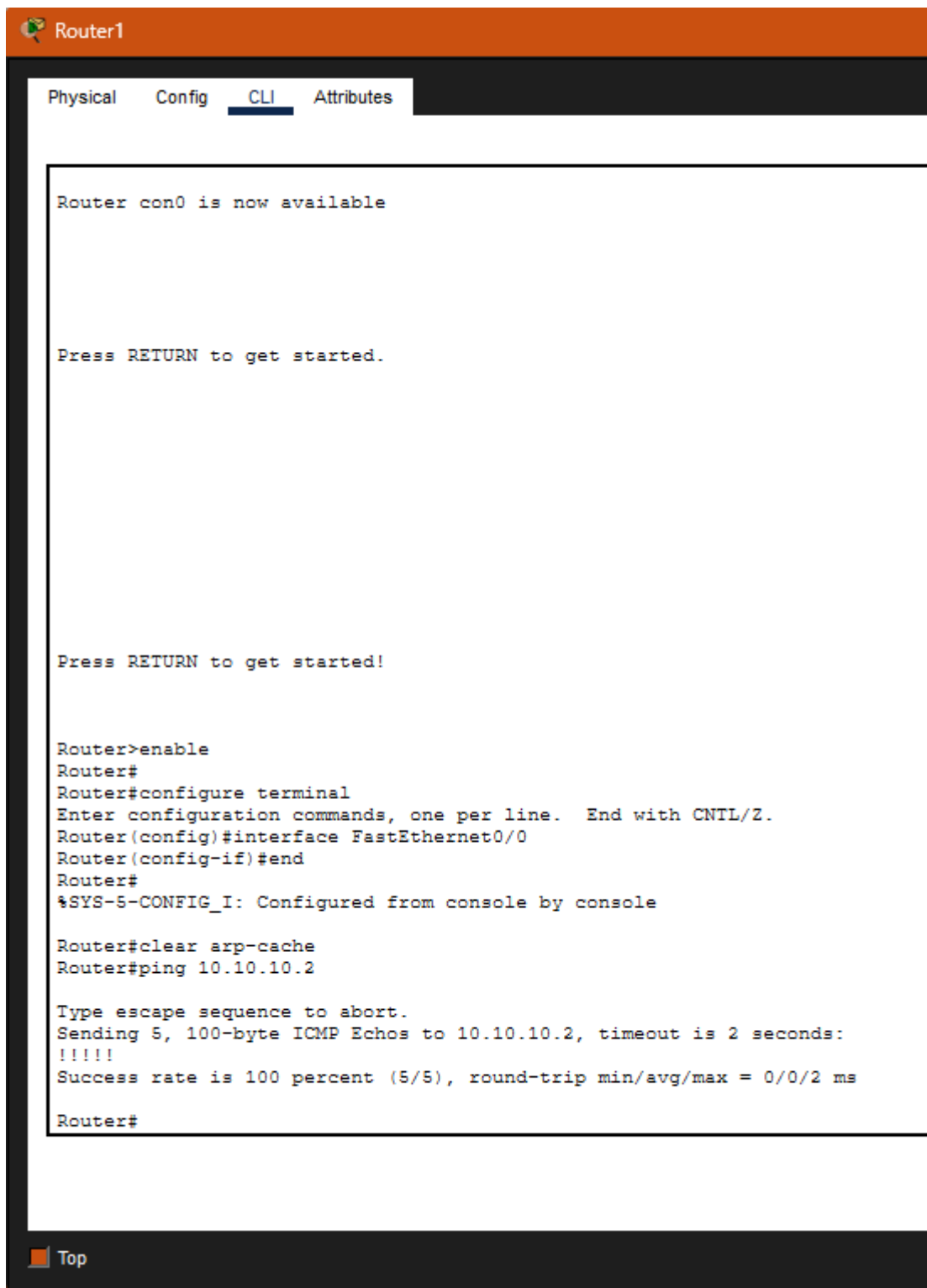
Router#ping 10.10.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Top

3.) screenshot of ARP clearing cache:



4.)Screenshot of show arp:

```
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:  
!!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/2 ms
```

```
Router#show arp  
Protocol Address Age (min) Hardware Addr Type Interface  
Internet 10.10.10.1 - 0003.E473.5C01 ARPA FastEthernet0/0  
Internet 10.10.10.2 0 0090.0CE6.B501 ARPA FastEthernet0/0  
Router#
```

 Top

```
Router#show arp  
Protocol Address Age (min) Hardware Addr Type Interface  
Internet 10.10.10.1 8 0003.E473.5C01 ARPA FastEthernet0/0  
Internet 10.10.10.2 - 0090.0CE6.B501 ARPA FastEthernet0/0  
Internet 10.10.20.1 4 0090.2B55.0501 ARPA FastEthernet0/1  
Internet 10.10.20.2 - 0090.0CE6.B502 ARPA FastEthernet0/1  
Router#
```

 Top

```
Router#show arp  
Protocol Address Age (min) Hardware Addr Type Interface  
Internet 10.10.20.1 - 0090.2B55.0501 ARPA FastEthernet0/0  
Internet 10.10.20.2 5 0090.0CE6.B502 ARPA FastEthernet0/0  
Router#
```

 Top

Reflection Questions:

20.) An ARP entry maps an IP address to a MAC address

21.) A router is in need of ARP because the route tells it where the packet should go at layer 3.